

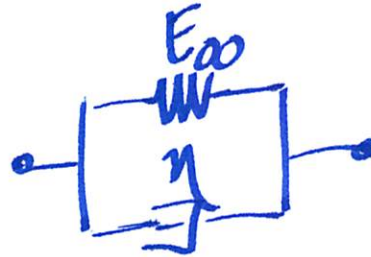
Viscoelasticity

1/

Maxwell Model

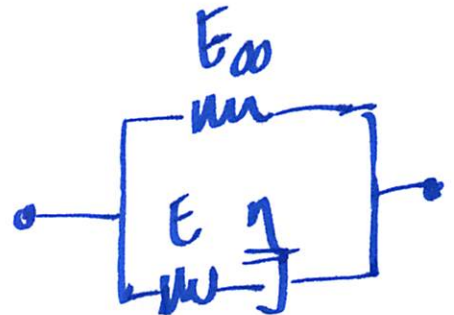


Kelvin-Voigt Model



Standard Model
(Zener Model)

└ Maxwell Representation



└ Kelvin Representation



For reasons:
From physical observations

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$$E > 0, E_{\infty} > 0, E_0 > 0, \eta > 0$$

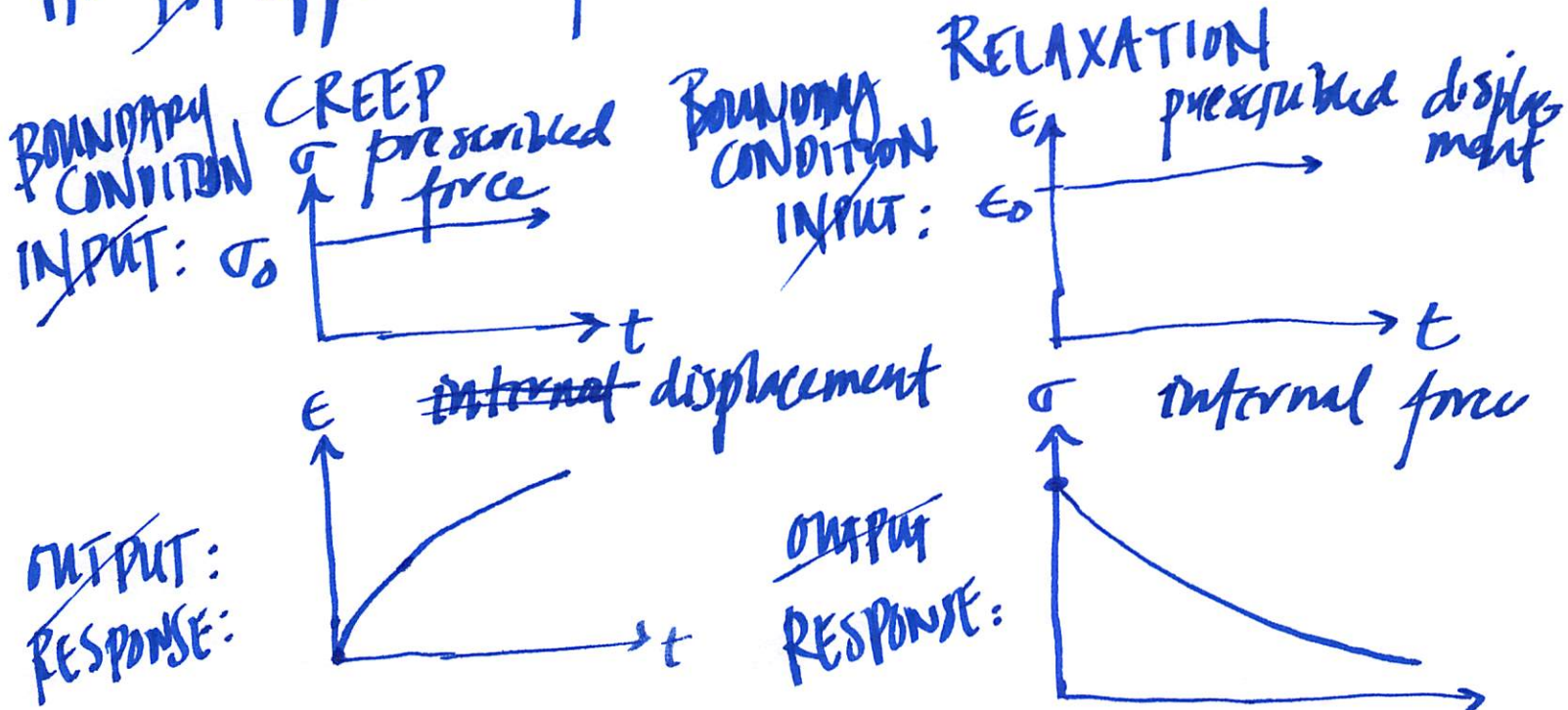
We have not yet specified how end points

move.

End points will move either by applied force σ_0 or applied displacement ϵ_0 .

For applied force, we can observe creep.

For applied displacements, we observe relaxation.



Time evolution.

- Let $[0, T] \subset (\mathbb{R}^+ \cup \emptyset)$ be the time interval of interest.
- Let the extended time interval be $(-\infty, T]$
- Let an internal variable $\alpha: (-\infty, T] \rightarrow \mathbb{R}$ be interpreted as the inelastic strain $\alpha(t)$ in the dashpot.
- Let a linear constitutive ~~law~~ law ^{relate} describe the inelastic strain rate $\dot{\alpha}(t)$ to the viscous stress σ_v as

$$\sigma_v = \eta \dot{\alpha}.$$

Note that η will have units of $\text{Ns} \cdot \text{m}^{-2}$.

Ex: Consider the Standard Model, Maxwell Representation, subjected to external force σ_0 ^{displacement} and responding with creep causing causes a ^{relaxation} creep response. The parallel construction of the model results in ~~an additive internal force~~ the total internal force to be the addition of each of the two branches.

Balance of linear momenta requires

$$\underbrace{\sigma_E + \sigma_V}_{\text{internal force}} = \underbrace{\sigma_0}_{\text{external force.}}$$

~~or equivalently,~~

$$\underline{E\epsilon + \eta}$$

where σ_E is ^{the} elastic contribution
 σ_V is ^{the} ~~metastatic~~ viscous (inelastic) contribution.

Since Let the total (elastic plus inelastic) strain in the element be $\epsilon(t)$. ϵ_0 .
 Fully elastic upper; Delastic + inelastic lower

The ^{internal} force in the top element is $E\alpha(t)$. 5/
 The ^{internal} force in the bottom element is

$$\eta \dot{\alpha} + E(\epsilon - \alpha).$$

dashpot

spring strain in E
 spring E .

Since the force in the ~~lower~~ across the
 spring E and ~~dashpot~~ dashpot is always
 identical ~~for~~ from equilibrium,

$$\eta \dot{\alpha} = E(\epsilon - \alpha).$$

Let $\tau \triangleq \eta/E$ be defined as the relaxation
 time. ^{Note τ has units of seconds.} Then,

$$\dot{\alpha} + \frac{1}{\tau} \alpha = \frac{1}{\tau} \epsilon$$

is the evolution equation with the
 initial condition

$$\alpha(t \rightarrow -\infty) = 0.$$

In the limit as $t \rightarrow -\infty$, $\alpha \rightarrow 0$, then 6/
 $\alpha = 0$ and the force in the lower branch
 σ is $E\epsilon$, at which time

$$\underbrace{(E_{\infty} + E)}_{\text{internal}} \epsilon = \underbrace{\sigma_0}_{\text{external}}$$

$$E_0 \epsilon = \sigma_0$$

as $E_0 \triangleq E_{\infty} + E$ is defined as the initial modulus.

The evolution equation $\dot{\alpha} + \frac{1}{\tau} \alpha = \frac{1}{\tau} \epsilon$ gives rise
to the integration factor $\exp(t/\tau)$,

thus
$$\frac{d}{dt} (\exp(t/\tau) \alpha(t)) = \frac{\exp(t/\tau) \epsilon(t)}{\tau}$$

as
$$\cancel{\exp(t/\tau)} \frac{1}{\tau} \alpha(t) + \cancel{\exp(t/\tau)} \dot{\alpha}(t) = \cancel{\exp(t/\tau)} \frac{1}{\tau} \epsilon(t)$$

$$\dot{\alpha} + \frac{1}{\tau} \alpha = \frac{1}{\tau} \epsilon. \quad \checkmark$$

Integration of $\frac{d}{dt}[(\cdot)\alpha] = (\cdot)\frac{e}{\tau}$ 7/

gives

$$\alpha(t) = \frac{1}{\tau} \int_{-\infty}^t \underbrace{\exp(-(t-s)/\tau)}_{dv} \underbrace{e(s)}_u ds$$

Integrating by parts

$$\int_a^b u dv = uv \Big|_a^b - \int_a^b v du$$

gives

$$\alpha(t) = \exp(-(t-s)/\tau) e(s) \Big|_{-\infty}^t - \int_{-\infty}^t \exp(-(t-s)/\tau) \dot{e}(s) ds$$

$$= (1)(e(t) - (0)(\underbrace{e(-\infty)}_{\rightarrow 0}))_{\rightarrow 0} \quad e(t) \rightarrow 0 \text{ as } t \rightarrow -\infty$$

$$\alpha(t) = e(t) - \int_{-\infty}^t \exp(-(t-s)/\tau) \dot{e}(s) ds$$

Now since internal free evolution

$$\sigma(t) = E_{\infty} e + E(e - \alpha)$$

$$= E_{\infty} e + E \int_{-\infty}^t \exp(-(t-s)/\tau) \dot{e}(s) ds$$